

INDIVIDUAL TECHNICAL REPORT

on

POOR WASTE SEGREGATION

Sustainable Development Goals Addressed: SDG 11, SDG 12 and SDG 13

Proposed Individual Solution:

*“EcoSort AI” – An AI-Powered Web Application for Real-Time Waste Segregation
Guidance and Community Compliance Monitoring*

Submitted by:

Name : Nikitha Praveen
Class : S3 CSE delta
Roll No : 06
UID : U2503215

Abstract

Municipal solid waste management remains one of the most persistent urban sustainability challenges faced by developing economies, and poor waste segregation at source is a central cause of its inefficiency. Government data indicates that Indian cities generate over sixty million tonnes of municipal solid waste every year, yet only a minority of this waste is scientifically segregated before collection, resulting in contaminated recyclables, higher landfill loads, and elevated greenhouse gas emissions from mixed organic waste. This report examines the environmental, social and regulatory dimensions of poor waste segregation and proposes an original engineering/IT-based intervention, EcoSort AI, an artificial-intelligence-powered web application that guides households, campuses and municipal workers to correctly classify waste in real time using image recognition, verifies disposal through QR-coded smart bins, and drives behavioural change through a community compliance dashboard and gamified incentives. The report evaluates the proposal against Sustainable Development Goals 11, 12 and 13, discusses its alignment with ISO 14001 environmental management principles, presents a qualitative life-cycle comparison of segregated versus mixed waste streams, and analyses technical, economic, operational and social feasibility. The report concludes that a low-cost, scalable digital nudge combined with physical verification infrastructure can meaningfully improve source-segregation compliance and support circular-economy outcomes.

Introduction

Rapid urbanisation, rising consumption and changing lifestyles have caused a steep increase in the volume and complexity of municipal solid waste (MSW) generated across the world. In India alone, urban centres are estimated to generate between 130,000 and 150,000 metric tonnes of solid waste every single day, a figure projected to more than double by 2031 as cities continue to expand. A large share of this waste is never scientifically segregated at the household or institutional level; official assessments indicate that only about one-third of the municipal waste generated annually is properly segregated before it enters the collection and treatment chain. The remainder is collected as a mixed, contaminated stream that is far more difficult, expensive and environmentally damaging to process.

Waste segregation at source - the practice of separating waste into categories such as wet/biodegradable, dry/recyclable, and hazardous/e-waste at the point of generation - is widely recognised as the single most cost-effective intervention in the entire solid waste management value chain. When segregation fails, biodegradable waste contaminates recyclable material, recyclers reject or downgrade otherwise valuable materials, and a larger proportion of total waste is diverted to landfills, where it decomposes anaerobically and releases methane, a greenhouse gas with a global warming potential many times greater than carbon dioxide over a 20-year horizon.

India's regulatory framework, particularly the Solid Waste Management Rules, 2016 and their successor policy updates, legally mandates segregation of waste into at least three streams by all waste generators, and empowers urban local bodies to levy differential fees on unsegregated waste. However, enforcement has been inconsistent, public awareness remains low, and there is a systemic absence of real-time feedback that tells a citizen, at the moment of disposal, whether they have segregated correctly. This gap between regulatory intent and behavioural reality is precisely the opportunity this report addresses.

This report has been prepared as an individual technical submission under the topic "Poor Waste Segregation", mapped primarily to Sustainable Development Goal 11 (Sustainable Cities and Communities) and Goal 12 (Responsible Consumption and Production), with secondary linkages to Goal 13 (Climate Action). The report is structured to first establish the problem and its sustainability dimensions, then to present an original, individually-conceived engineering/IT solution - EcoSort AI - followed by feasibility, environmental benefit and justification analyses, before concluding with recommendations.

Problem Statement

Despite regulatory mandates requiring source segregation of municipal solid waste, actual compliance at the household, institutional and commercial level remains poor. The problem manifests through several interlinked failure points:

- Lack of real-time guidance: Citizens are often unsure which category a given item belongs to (e.g., a soiled food container, a tetra pack, or a small e-waste item), and no immediate feedback exists at the point of disposal to correct mistakes.
- Inconsistent or absent bin infrastructure: Many residential blocks, campuses and public spaces still rely on a single common bin, or provide colour-coded bins without adequate signage or maintenance.
- Contamination of recyclables: Even where dry waste is separated, mixing of wet or hazardous items renders a large share of “recyclable” material unfit for processing, reducing the effective recycling rate far below the collected volume.
- Absence of accountability and data: Local bodies rarely have granular, bin-level or household-level visibility into where and how often segregation fails, making it difficult to target awareness campaigns or enforcement efficiently.
- Low behavioural incentive: Correct segregation currently offers no visible or immediate personal benefit to the individual generator, so compliance depends almost entirely on civic conscience or the threat of penalty.
- Risk to informal waste workers: Poorly segregated waste, especially when it contains hazardous or sharp material, directly endangers the health and safety of sanitation workers and informal waste-pickers who manually sort mixed waste downstream.

The core problem addressed by this report can therefore be stated as follows: households, campuses and small institutions lack an accessible, real-time, and engaging mechanism to correctly identify and verify waste segregation at the point of disposal, resulting in contaminated waste streams, avoidable landfill burden, and weakened progress toward circular-economy and climate goals. The scope of this report is limited to source segregation at the residential and institutional (e.g., college campus) level, since this is the stage at which the highest-leverage, lowest-cost interventions are possible.

Sustainability Analysis

1. Relevant Sustainable Development Goals (SDGs)

The problem of poor waste segregation and the proposed solution align most directly with three Sustainable Development Goals:

- **SDG 11 - Sustainable Cities and Communities**: Target 11.6 specifically calls for reducing the adverse per-capita environmental impact of cities, including through attention to municipal and other waste management. Improved source segregation directly reduces the volume of waste requiring landfilling within urban limits.

- **SDG 12 - Responsible Consumption and Production:** Target 12.5 calls for substantially reducing waste generation through prevention, reduction, recycling and reuse. Clean, well-segregated recyclable streams are a precondition for functioning recycling markets and a circular economy.
- **SDG 13 - Climate Action:** Diverting biodegradable waste away from anaerobic landfill decomposition reduces fugitive methane emissions, directly supporting climate mitigation targets.

Secondary linkages also exist with SDG 3 (Good Health and Well-being), since safer handling of segregated waste protects sanitation workers, and SDG 17 (Partnerships for the Goals), since the proposed digital platform is designed to connect citizens, resident associations and municipal bodies on a shared data platform.

2. Alignment with ISO 14001 (Environmental Management Systems)

ISO 14001:2015 provides a globally recognised framework for organisations - including residential societies, educational campuses and municipal departments - to establish, implement and continually improve an Environmental Management System (EMS) based on the Plan-Do-Check-Act (PDCA) cycle. Poor waste segregation directly undermines an organisation's ability to meet the “Aspect and Impact Identification” and “Operational Control” clauses of ISO 14001, since without segregated, quantifiable waste streams, an organisation cannot accurately measure its waste-related environmental aspects or demonstrate continual improvement.

The proposed EcoSort AI solution supports ISO 14001 alignment in the following ways:

- **Plan:** Baseline waste-composition data captured by the AI classifier helps an organisation identify significant environmental aspects related to waste.
- **Do:** QR-coded bin verification operationalises correct segregation as a controlled, auditable process rather than an informal practice.
- **Check:** The community compliance dashboard functions as a built-in monitoring and measurement tool, satisfying documentation requirements for internal audits.
- **Act:** Contamination alerts and trend data allow facility managers to take timely corrective action and set continual-improvement targets, exactly as ISO 14001 requires.

3. Life Cycle Perspective (LCA) and Environmental Impact Considerations

A full quantitative Life Cycle Assessment (per ISO 14040/14044) was outside the scope of an individual technical report, but a qualitative, literature-informed comparison of segregated versus mixed waste handling illustrates why source segregation is the highest-leverage intervention point in the waste life cycle, from generation through collection, transport, processing and final disposal. The table below summarises the expected directional impact.

Life-Cycle Stage	Impact under Poor / Mixed Segregation	Impact with Improved Source Segregation
Generation / Source	No differentiation of waste streams; hazardous items mixed with general waste	Streams separated at source; hazardous/e-waste isolated for safe handling
Collection & Transport	Single mixed-load vehicles; higher trip frequency and fuel use per tonne recovered	Optimised, category-wise collection reduces re-handling and vehicle trips
Processing / MRF	Low recovery yield; high rejection rate of contaminated recyclables	Higher recovery yield; cleaner recyclate suitable for higher-value recycling
Final Disposal	Larger landfill volume; anaerobic decomposition of organics generates methane	Reduced landfill inflow; organics diverted to composting/biomethanation

Note: The comparison above is qualitative and directional, drawn from published solid-waste-management literature and government status reports

Regarding Environmental Impact Assessment (EIA): while a citizen-facing digital application does not itself trigger statutory EIA requirements, the underlying waste-management infrastructure it supports (e.g., material recovery facilities, composting yards, or landfill sites) falls under India's EIA Notification framework when expanded or newly constructed. Improved segregation data generated by this solution can therefore also assist urban local bodies in the environmental impact documentation required for such facility planning.

Proposed Engineering/IT Solution: EcoSort AI

This report proposes EcoSort AI, an individually-conceived, AI-powered web application designed to close the real-time feedback gap identified in the problem statement. Unlike generic informational waste-sorting apps or static colour-coded signage, EcoSort AI combines computer-vision-based classification, physical verification through smart-bin QR tagging, and a gamified community dashboard into a single closed-loop system. A working front-end prototype has been developed to demonstrate the proposed user experience; representative screenshots are provided in the Appendix.

1. System Overview and Core Modules

- **AI Image-Based Waste Classifier:** A user photographs a waste item using a phone or campus kiosk camera; a lightweight convolutional neural network (transfer-learned on a MobileNetV2 backbone) classifies the item into wet, dry-recyclable, hazardous/e-waste or sanitary categories and returns the correct bin along with a one-line disposal tip, in under two seconds.
- **QR-Coded Smart-Bin Verification:** Every physical bin in a pilot zone carries a unique QR code. Scanning the code before disposal timestamps and geo-tags the event, cross-references it against the AI-classified category, and flags mismatches for review - turning an informal action into an auditable record.

- Community Compliance Dashboard: Aggregated, anonymised data is displayed on a public-facing dashboard showing block-wise or ward-wise segregation compliance rates, active bin counts, and contamination alerts, giving residents' welfare associations and facility managers the visibility needed to act.
- Gamified Incentive Layer: Verified correct disposals earn “Green Points” that accumulate on a leaderboard, which can be tied to small municipal or campus incentives (fee rebates, recognition certificates), converting a compliance obligation into a rewarding habit.
- Administrator Alert Module: Facility managers and municipal ward officers receive automated alerts when a bin shows a rising contamination trend, enabling targeted, evidence-based awareness intervention rather than blanket campaigns.

2. Indicative Technology Stack

- Front-end: HTML5, CSS3 and JavaScript (responsive web application, installable as a Progressive Web App for offline-tolerant use).
- AI/ML Layer: TensorFlow Lite / MobileNetV2 transfer-learning model for on-device or edge inference, trained on an open waste-classification image dataset augmented with local waste samples.
- Backend & Database: Lightweight REST API (Node.js/Flask) with a relational or Firebase database to store classification logs, bin events and compliance scores.
- QR Infrastructure: Standard QR code generation/scanning libraries printed on weatherproof bin labels; no specialised hardware required.
- Hosting: Cloud-hosted on a low-cost tier (e.g., a student/education cloud credit plan) sufficient for a pilot of a few hundred households or a single campus.

3. What Makes This Solution Unique

Several categories of existing interventions address poor segregation only partially: printed signage and awareness posters provide no real-time correction; purely informational mobile apps tell a user the rule but cannot verify that it was followed; and CCTV-based bin monitoring is costly and raises privacy concerns. EcoSort AI is distinguished by closing the entire loop - guidance (AI classification) → verification (QR-tagged disposal) → feedback (dashboard and alerts) → incentive (gamification) - within a single low-cost, camera-and-QR-based system that requires no new bin hardware, making it realistically deployable by a single student, resident association or campus green-club team as a pilot.

Feasibility Analysis

The feasibility of EcoSort AI has been assessed across five standard dimensions, summarised in the table below.

Dimension	Assessment
Technical	High. The required components — lightweight image-classification models, QR generation/scanning, and web dashboards — are mature, well-documented open-source technologies that do not require specialised or proprietary hardware.
Economic	Favourable for a pilot. Estimated one-time cost of approximately ₹15,000–₹25,000 for QR labels, basic cloud hosting for one year, and a low-cost smartphone/kiosk for a single residential block or campus zone; ongoing costs are limited to hosting and periodic model retraining.
Operational	Moderate. Requires initial resident/student onboarding, printed QR labels on existing bins, and a designated volunteer or facility staff member to review contamination alerts; no changes to existing collection vehicle routes are required at pilot scale.
Social	High. The gamified leaderboard and visible dashboard are designed to leverage social recognition and mild peer competition, both of which are well-established behavioural drivers for civic compliance.
Legal / Regulatory	Compliant by design. The solution directly supports compliance with the Solid Waste Management Rules, 2016 (source segregation mandate) and can be operated with anonymised or aggregated data to respect resident data-privacy expectations.

Overall, the solution is assessed as feasible for a scoped pilot deployment (a single residential block, hostel, or academic department) within one academic semester, using freely available datasets and open-source tooling, with a realistic pathway to scale if the group implementation phase selects this proposal.

Expected Environmental Benefits

- **Reduced landfill burden:** Higher-accuracy source segregation reduces the volume of biodegradable and recyclable material that is otherwise land-filled as mixed waste.
- **Lower greenhouse gas emissions:** Diverting organic waste toward composting or biomethanation reduces anaerobic methane generation at disposal sites, contributing to SDG 13 targets.
- **Higher-value recycling:** Cleaner, uncontaminated dry-waste streams command better prices and higher recovery yields in the recycling market, strengthening circular-economy outcomes.
- **Improved worker safety:** Verified segregation of hazardous and sanitary waste reduces the exposure of sanitation workers and informal waste-pickers to unsafe material.
- **Behavioural and awareness impact:** The gamified dashboard is expected to produce a measurable, sustained increase in resident/student awareness and compliance, an outcome that is difficult to achieve through signage alone.
- **Data for policy:** Ward- or block-level compliance data can help municipal bodies target awareness campaigns and enforcement more precisely, improving the cost-effectiveness of public spending on waste management.

Justification of the Proposed Solution

The proposed EcoSort AI solution is justified as the preferred individual response to the assigned problem for several reasons. First, it directly targets the root cause identified in the problem statement - the absence of real-time, point-of-disposal feedback - rather than only reinforcing awareness, which existing signage-based interventions already attempt with limited success. Second, it is technically and economically feasible at a scale a single student or small group can realistically pilot, unlike infrastructure-heavy alternatives such as installing sensor-based smart bins or CCTV surveillance networks, which involve significantly higher capital cost and raise privacy concerns.

Third, the solution is measurable: every disposal event generates a data point, allowing the impact of the intervention to be tracked and reported quantitatively (e.g., percentage compliance before and after pilot deployment), which strengthens both the sustainability case and the ISO 14001 continual-improvement narrative developed in Section 3.2. Fourth, the solution is inherently scalable - the same AI model and dashboard architecture that serves a single hostel block can, with additional training data and bin coverage, be extended to an entire ward or municipal zone, aligning with national digital-governance initiatives such as the Swachh Bharat Mission-Urban and various Smart Cities Mission digital platforms.

Finally, compared to purely policy-based alternatives such as stricter fines or mandatory manual audits, EcoSort AI relies on positive reinforcement (gamification) rather than punitive enforcement, which is both more socially acceptable for a resident-facing pilot and better aligned with the participatory, community-driven spirit of Sustainable Development Goal 11.

Conclusion

Poor waste segregation at source remains a significant obstacle to achieving India's solid-waste-management and climate goals, despite a clear regulatory mandate under the Solid Waste Management Rules, 2016. This report has examined the problem through the lens of Sustainable Development Goals 11, 12 and 13, situated it within an ISO 14001 environmental-management framework, and presented a qualitative life-cycle argument for why source segregation is the highest-leverage intervention point available. In response, this report proposes EcoSort AI, an individually-developed AI-powered web application that closes the guidance-verification-feedback-incentive loop using image-based waste classification, QR-coded bin verification, and a gamified community compliance dashboard. The feasibility analysis indicates that a scoped pilot is achievable within realistic student-project constraints, and the expected environmental benefits - reduced landfill load, lower methane emissions, cleaner recyclate and safer working conditions for sanitation workers - justify its consideration for the group implementation phase of this project.

References

- [1] Ministry of Housing and Urban Affairs (MoHUA), Government of India, “Solid Waste Management Rules, 2016.”
- [2] Bisht, G., “Solid Waste Management Rules, 2026: Examining India’s Implementation Readiness,” IMPRI Policy Update, IMPRI – Impact and Policy Research Institute, New Delhi.
- [3] NITI Aayog / Centre for Science and Environment, Foreword to the Report on Municipal Solid Waste Management in Indian Cities, 2021.
- [4] Central Pollution Control Board (CPCB), Protocol for Evaluation of Technology for Waste Management, cpcb.nic.in.
- [5] Swachh Bharat Mission – Urban 2.0, Ministry of Housing and Urban Affairs, Government of India.
- [6] United Nations, “The 17 Sustainable Development Goals – Goal 11, Goal 12 and Goal 13,” sdgs.un.org.
- [7] International Organization for Standardization, ISO 14001:2015 – Environmental Management Systems – Requirements with Guidance for Use.
- [8] International Organization for Standardization, ISO 14040:2006 – Environmental Management – Life Cycle Assessment – Principles and Framework.
- [9] Google / TensorFlow, “TensorFlow Lite and MobileNetV2 Model Documentation,” tensorflow.org.
- [10] Ministry of Environment, Forest and Climate Change, Government of India, Environmental Impact Assessment (EIA) Notification framework, moef.gov.in.

Appendix – Screenshots of Website Pages

The EcoSort AI web application described in Section 4 is proposed as part of this individual technical report and has not yet been developed as a functioning website at the time of this submission. As such, no screenshots are included here. Once the solution advances to the prototyping/implementation stage (including, potentially, the group implementation phase of this project), this appendix will be updated with actual screenshots of the home page, the AI waste-scanner interface, and the community compliance dashboard, each showing the live classification and reporting functionality described in Section 4.

Planned pages to be captured at that stage:

1. Home / Landing Page - introducing EcoSort AI and its core modules.
2. AI Waste Scanner Page - showing image upload and real-time classification result.
3. Community Compliance Dashboard - showing block-wise segregation compliance and the leaderboard.